



BUSINESS BRIEF

Ndarama Grid Business Overview

A commercial view of a mine-site operations platform



PitGuard CAM-02 image shown inside the application. Demonstration imagery, not a live camera feed.

MineTech Innovation Challenge Zimbabwe 2026

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Executive view

Ndarama Grid brings mine safety workflows, mineral custody records, offline communications and day-to-day planning into one local application. The current product is a working software prototype with a local Edge Hub, a 12-window desktop interface and an Android companion. Its strongest near-term commercial case is a controlled site pilot that tests whether small and mid-sized operators will pay for a simpler, locally managed digital workflow.

The commercial case remains unproven. The materials supplied for this overview do not include customer interviews, signed pilots, deployed-device results, revenue, unit costs or a validated market size. The opportunity is therefore framed as a set of hypotheses to validate, rather than as a priced investment or forecast.

Business problem and product response

Mine teams may work across disconnected shifts and underground areas where cloud-first systems are difficult to use. Operators also need to keep worker alerts, sensor observations, custody records, maintenance and planning connected to the people making decisions. Ndarama Grid proposes a site-based Edge Hub that can continue locally and sync when a network is available.

Initial buyer and user hypotheses

Potential buyer	Operational reason to evaluate	Validation question
ASM cooperative or mine operator	Coordinate safety records, roster checks, assets and site-level data.	Who owns the budget and who is accountable for safe operation?
Mineral buying centre or traceability partner	Check parcel identity and custody history at handover.	Will counterparties accept the records and verification process?
Mine services or equipment partner	Deploy and maintain site hardware with a software workflow.	Can the partner support devices, calibration and response obligations?

Product and deployment model

The desktop application and local Edge Hub provide the current software layer. SQLite stores site records locally. Android can connect to the PC Hub over the same private Wi-Fi using the address and port shown in desktop Settings. The phone camera can decode a parcel QR value and submit that value to the Hub for a ledger lookup and integrity check.

Product area	Current software proposition	Field dependency
PitGuard	Safety view for methane, carbon monoxide, dust, seismic readings, PPE status and incidents.	Sensor, camera and alert pathways require site hardware, calibration and end-to-end validation.
OreChain	Parcel records, QR lookup, custody history and hash/signature verification.	Buying-centre acceptance, tag process and real transaction records must be proven.
Mesh	Store-and-forward queue concept with life-safety priority over bulk sync.	LoRa radios, underground coverage, power and failover need field testing.
Copilot and operations	Local or provider-based advisory, planning, asset and workforce screens.	Advice needs human review; data quality and site-specific workflow fit remain open.

Deployment hypothesis

Install the Hub on a low-cost PC or suitable site computer. Configure the site profile and user roles, add or connect validated devices, then train supervisors and records staff. Keep the system on a trusted site network, apply an API token where needed, and define backup, update and support ownership before live use.

A pilot should begin with one site and a small set of users. It should test offline behavior, recovery after power or network interruption, worker alert delivery, custody handover and daily use. The pilot should measure work completed and failure cases, not just screen activity.

Commercial pathways to test

These are candidate models for customer discovery. The project files do not establish pricing, gross margin or a preferred model.

Candidate revenue	What a customer could buy	Evidence required
Site software subscription or licence	Access to the Hub and operator interface, with updates and support.	Willingness to pay, renewal terms, support costs and offline licence handling.
Deployment and training	Site configuration, onboarding, integrations and operating procedures.	Time to install, training effort and repeatability across sites.
Hardware and maintenance	Validated tags, sensor pods, network equipment or maintenance through qualified partners.	Bill of materials, certifications, calibration cycle, warranties and service coverage.
Traceability services	Parcel onboarding, verification or reporting for approved supply-chain participants.	Adoption at both ends of the chain, data governance and buyer acceptance.

Market entry sequence

- Interview operators, worker representatives and buying centres to select one costly, frequent workflow.
- Agree one pilot site, an operational owner and a baseline for current time, completeness and exception handling.
- Run the software in supervised mode while comparing every critical result against existing procedures.
- Add hardware only after the interface, alert path, servicing plan and safety responsibilities are specified.
- Convert successful pilot work into a repeatable deployment package, support offer and buyer-approved price test.

Investment case and validation plan

For a prospective investor or implementation partner, the immediate question is whether the product can move from a capable prototype to a reliable site service with a clear payer. An initial funding or partnership decision should be tied to evidence gates and a named pilot, not to an assumed national market share.

Gate	Evidence to produce	Decision enabled
Customer discovery	Named buyer, problem frequency, present workaround, budget route and procurement steps.	Select a beachhead use case and pilot partner.
Controlled software pilot	User adoption, record completeness, uptime, offline recovery and measured time saved.	Decide whether the workflow merits paid deployment.
Hardware validation	Coverage tests, calibrated readings, alert latency, battery life and maintenance records.	Decide which hardware can be safely offered and supported.
Commercial repeatability	Signed terms, deployment cost, support demand, renewal signal and partner responsibilities.	Decide whether to scale sales and capital commitment.

Key diligence questions

- Which buyer controls the budget, and what operational result will justify paying?
- Which screens use live hardware today, and which currently show demonstration or simulated values?
- What legal, mine-safety, radio and data-protection reviews apply before deployment?
- Who calibrates, repairs and replaces devices, and how quickly can a site regain service?
- How are user access, backups, signing keys, security updates and incident response managed?

Readiness statement

Ndarama Grid has a substantial software prototype and conceptual hardware material. The available evidence does not demonstrate certified sensing hardware, validated underground radio coverage, regulatory approval, paid customer use or audited financial results. Treat safety outputs as decision support until a qualified site programme validates the complete system.

Source basis: repository README, Edge Hub and React application, Android companion, product concept drawings and documentation available in the project workspace. Product screens can contain demonstration data. No market statistics or financial forecasts were supplied.